



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to BRIGHT AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

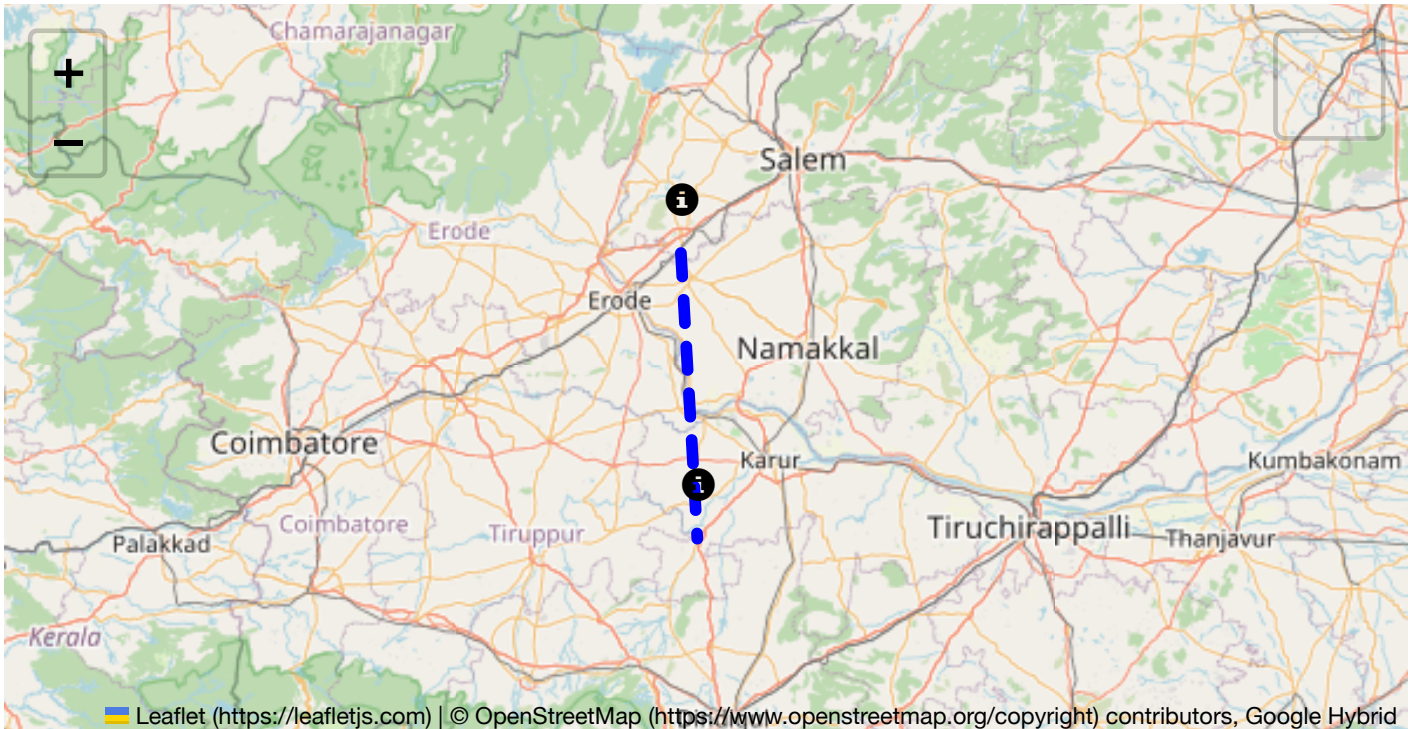
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 91.40 km
Estimated Duration: 1.7 hours
Adjusted Duration (Heavy Vehicle): 2.1 hours

Start: (11.4381, 77.8734)
End: (10.7817699, 77.911869)



Welcome to the Journey Risk Management Study

Overview of the Route Map

The route from Sangagiri (CVQF+23W) to Kharedipatti (QWM7+357) is approximately 91.40 kilometers and typically takes around 1.67 hours for heavy vehicles. Starting from Sangagiri, you head southeast towards NH44, which is one of the primary arteries, allowing efficient connectivity. From NH44, the route may continue along major state highways leading to Kharedipatti. This route passes through both rural and semi-urban areas.

Typical Weather Conditions and Potential Weather-Related Hazards

Southern India, including Tamil Nadu, generally experiences a hot climate, with temperatures peaking in summer. The pre-monsoon and monsoon seasons (June to September) might bring heavy rains, causing potential hazards such as waterlogging and reduced road traction. In the monsoon, fog and reduced visibility can pose additional risks.

Analysis of Traffic Patterns

Traffic is likely to be denser near urban centers and townships that the route bypasses or intersects with, particularly Erode and Karur districts. Peak hours generally align with local commuting times, typically from 8:00 to 10:00 AM and 5:00 to 7:00 PM. The busiest stretches are expected near intersections with other state and national highways.

Assessment of Road Quality and Infrastructure

The road quality along NH44 is typically well-maintained, with a few stretches possibly undergoing repair due to regular wear and tear or monsoon damage. Rural and secondary roads linking the main highway to

Kharedipatti may be narrower and less maintained. Watch out for uneven surfaces, especially after heavy rainfall seasons.

Suggestions for Alternative Routes for Emergencies

Alternatives include state highways that run parallel to NH44, though they may increase travel time slightly. Local roads provide quick exits but potentially lack infrastructure for heavy vehicle support. Use GPS navigation to find the nearest bypass routes in emergencies.

Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu has strict guidelines for transporting hazardous materials, mandating proper documentation and compliance with specific vehicular permits. Transporters must adhere to designated routes and avoid sensitive zones, and night transportation is often restricted or requires prior approval.

Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

Historically, incidents on this route are rare but not absent. The primary causes usually involve human error, improper loading, or vehicle maintenance failures rather than route-based issues. Nonetheless, always ensure compliance with safety standards.

Environmental Considerations and Sensitive Areas

The route may pass adjacent to agricultural lands and water bodies. Care should be taken to prevent any spills or contaminations that might impact these environments. Follow local ecological regulations to minimize potential disturbances.

Analysis of Communication Coverage

Although the route primarily spans an inhabited region with decent cellular coverage, some rural segments or areas in hilly terrain may experience weaker signals. Carriers like Airtel, Jio, and Vodafone are best suited for the region. Utilize offline maps as a precaution.

Estimated Emergency Response Times

Response times vary widely based on proximity to Krishna giri, Erode, and Karur town centers, where facilities are more robust. Rural areas can see delays of up to an hour for emergency services. Ensure availability and testing of emergency communication devices on board.

Overall Summary of Risk Assessment

This route is relatively safe for transportation, given proper adherence to safety measures and regulations. The principal risks arise from weather conditions, infrastructure variations, and unexpected traffic congestion. Adequate preparation and familiarity with alternative paths and communication methods are crucial for ensuring a smooth journey. It is recommended to perform routine checks on the vehicle, particularly regarding load security and mechanical health before departure. Additionally, continuous monitoring of weather reports and traffic updates will aid in mitigating unforeseen challenges.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16764, 78.01565	30 KM/Hr
8	Turn	High	11.14545, 78.02208	15 KM/Hr
9	Turn	High	11.14480, 78.02291	15 KM/Hr
10	Turn	Medium	11.10102, 78.00572	30 KM/Hr
11	Turn	High	10.79637, 77.92684	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Abishek Ortho Center	10.9628788, 78.0600715	30 km/h	Medium
19	hospital	Abishek Ortho Center, Karur	10.9624485, 78.059568	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
20	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37862, 77.89488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.79637, 77.92684

